XMAS DEV 2023

MONITORA LA FABBRICA DI BISCOTTI PER BABBO NATALE CON I SERVIZI GRATUITI AZURE

Nicola Paro

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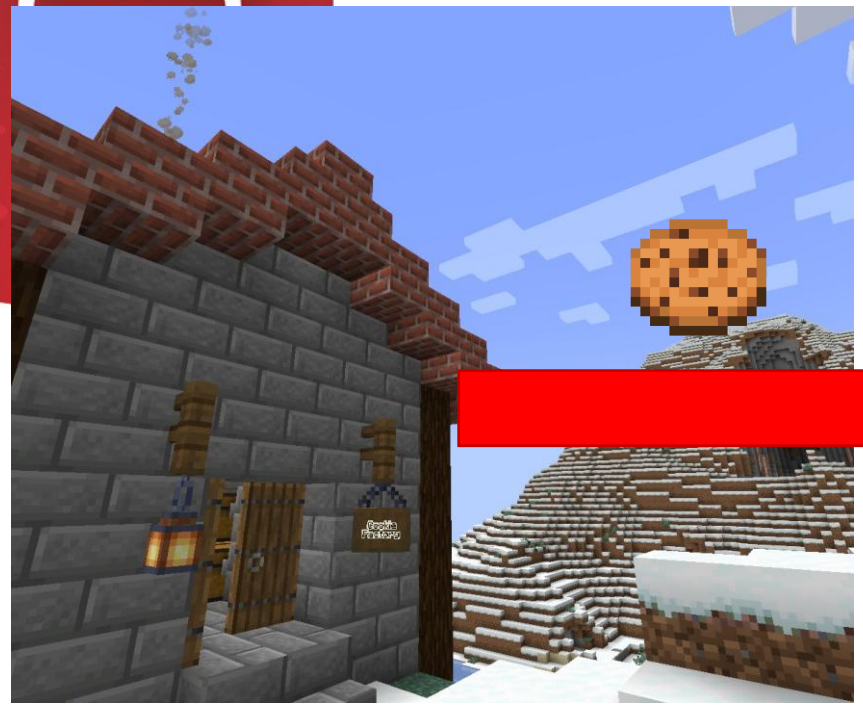
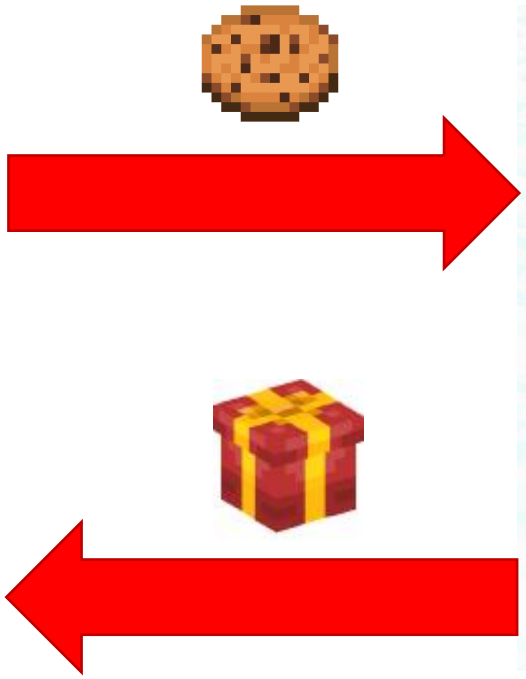
A VIRTUAL LIFE STORY

Il Natale è alle porte...

A VIRTUAL LIFE STORY

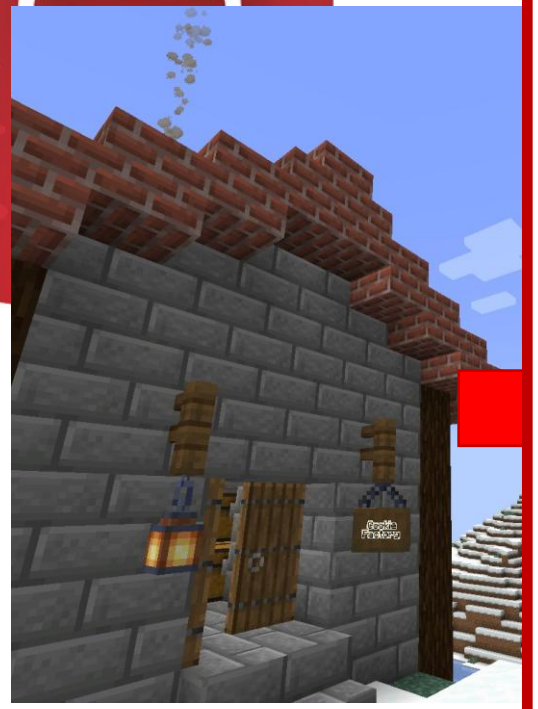
Il Natale è alle porte...





A VIRTUAL LIFE STORY

Il Natale è alle porte…



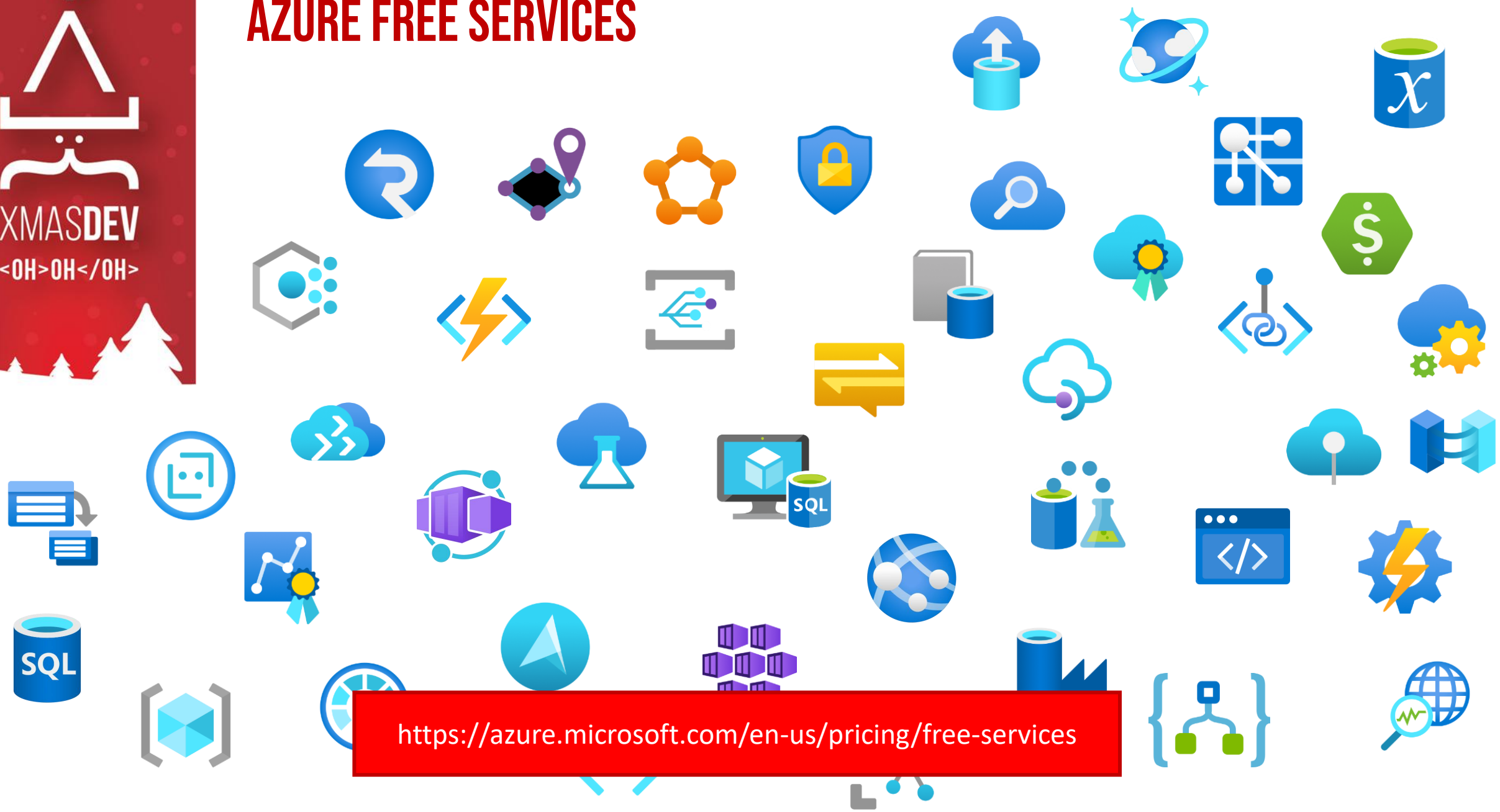
PROBLEMI

(o Opportunità?)

- Se finisce il cioccolato, la produzione si ferma
- La produzione ogni tanto si inceppa ed è richiesto un intervento manuale per ripristinarla
- Nessuno si accorge dei problemi, perché non sono notificati

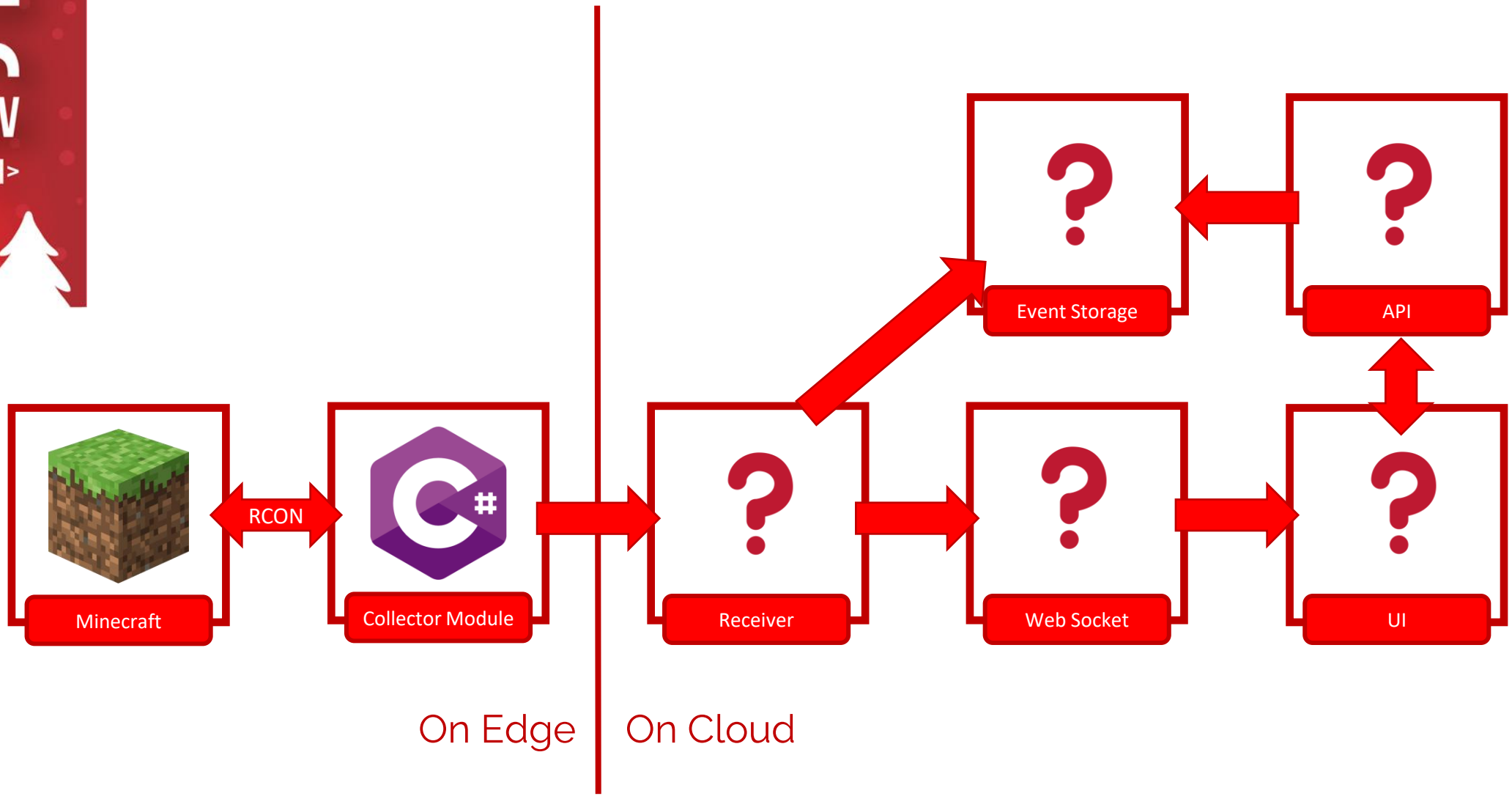
- Necessaria una soluzione di monitoraggio, ma Azure non si può pagare con i biscotti

AZURE FREE SERVICES

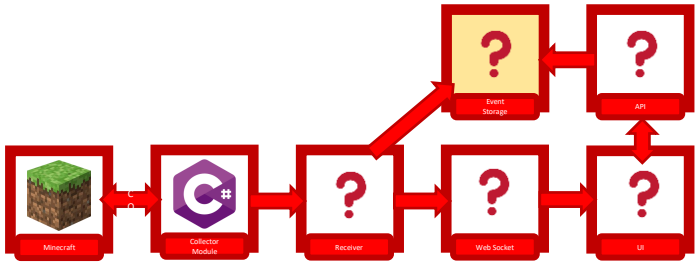


https://azure.microsoft.com/en-us/pricing/free-services

MONITORING PLATFORM OVERVIEW

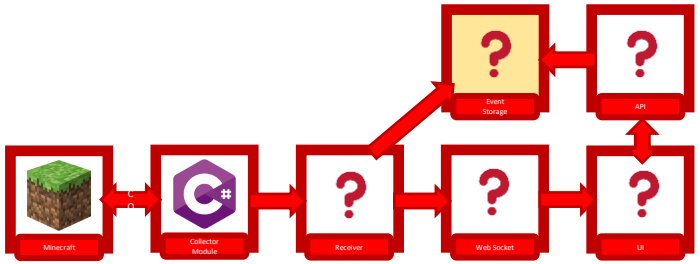


MONITORING PLATFORM – EVENT STORAGE



- Deve persistere gli eventi dell'impianto produttivo per poterli successivamente interrogare
- Deve permettere analitiche in modo semplice

COSMOS DB

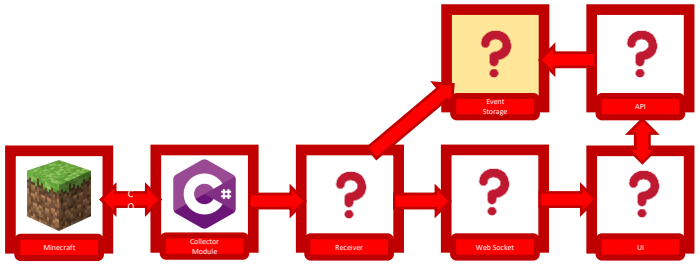


- Database gestito sia NoSQL che relazionale
- Globalmente distribuito, altamente responsive e sempre online
- Puoi usare l'API che preferisci

NoSQL	MongoDB	Apache Cassandra
Apache Gremlin	Table	PostgreSQL

- Free Tier: Max throughput 1000 RU/s + 25 GB storage

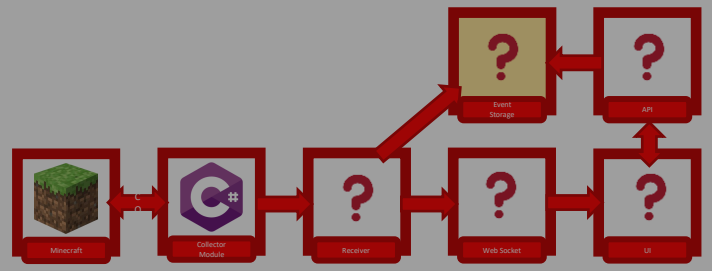
AZURE SQL



- Database relazionale totalmente gestito
- Costriuito sullo stesso engine di SQL Server

- Free Tier con 100,000 secondi * vCore e 32GB di storage limit al mese per sempre
  - Puoi scegliere di metterlo in auto-pause quando il limite è raggiunto
  - Oppure continuare ad utilizzarlo e pagare solamente per la parte extra

AZURE SQL DB – FREE TIER



Basics | Networking | Security | Additional settings

Cost summary

General Purpose (GP_Gen5_2)

Cost per **vCore** (in EUR) [1]	114.41
vCores selected	x 2
Cost per **GB** (in EUR)	0.13
Max storage selected (in GB)	x 41.6
ESTIMATED COST / MONTH	234.16 EUR

NOTES

[1] The dev/test discount has been automatically applied for your selected subscription. Learn more

Cost summary

General Purpose (GP_S_Gen5_2)

Cost per **GB** (in EUR)	0.00
Max storage selected (in GB)	x 41.6
First 32 GB storage free	
First 100,000 vCore seconds free	
Overage billing [1]	Disabled
ESTIMATED STORAGE COST / MONTH	0.00 EUR
COMPUTE COST / VCORE SECOND [2]	0.000000 EUR

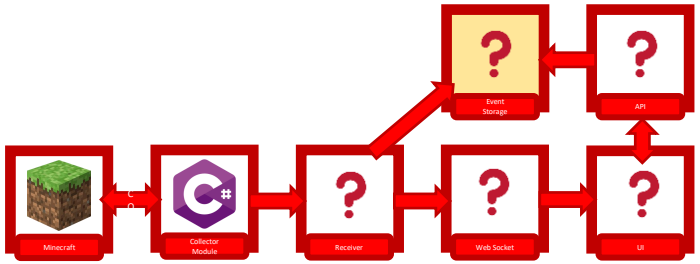
[1] There will be no charges for usage within the free limits. The database will be paused automatically when the free limits are reached.

[2] Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. Learn more about serverless billing

Create new

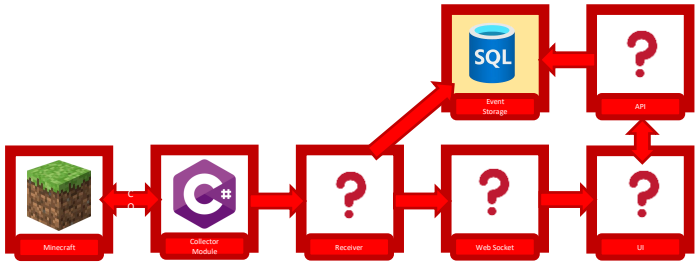
MONITORING PLATFORM – EVENT STORAGE

Cosmos o SQL ?

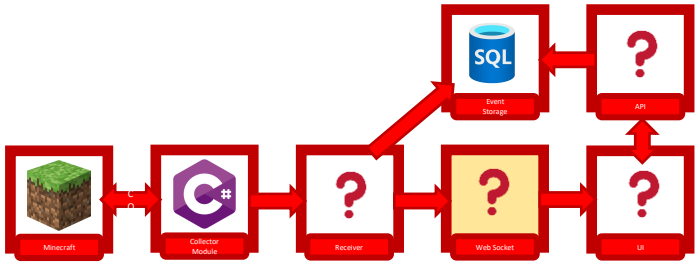


MONITORING PLATFORM – EVENT STORAGE

Cosmos o SQL ?

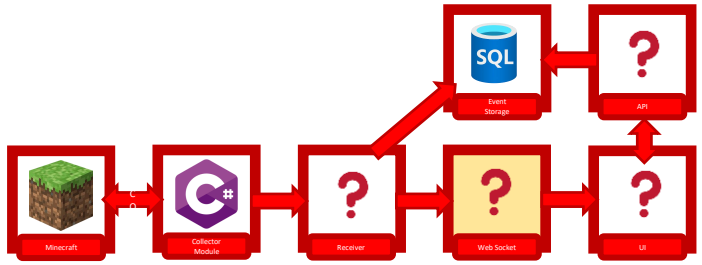


MONITORING PLATFORM – WEB SOCKET



- Deve notificare gli eventi dell'impianto al frontent per garantire un rapido intervento

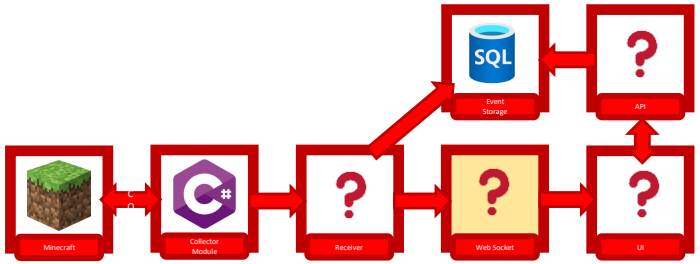
AZURE SIGNALR & WEB PUBSUB



Pensati per applicazioni che richiedono dati in tempo reale

High frequency updates	Live dashboards e monitoring	Cross-platform live chat	Real-time location on map	Real-time targeted ads
Collaborative apps	Push notifications	Real-time broadcasting	IoT	Automation

SIGNALR VS AZURE WEB PUBSUB

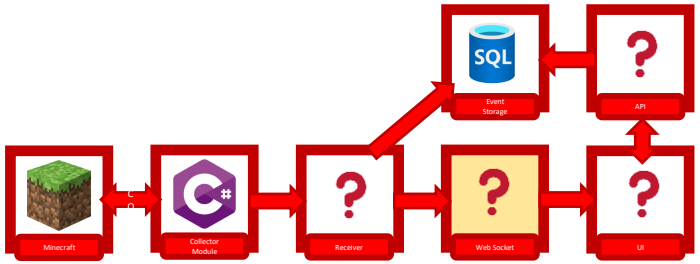


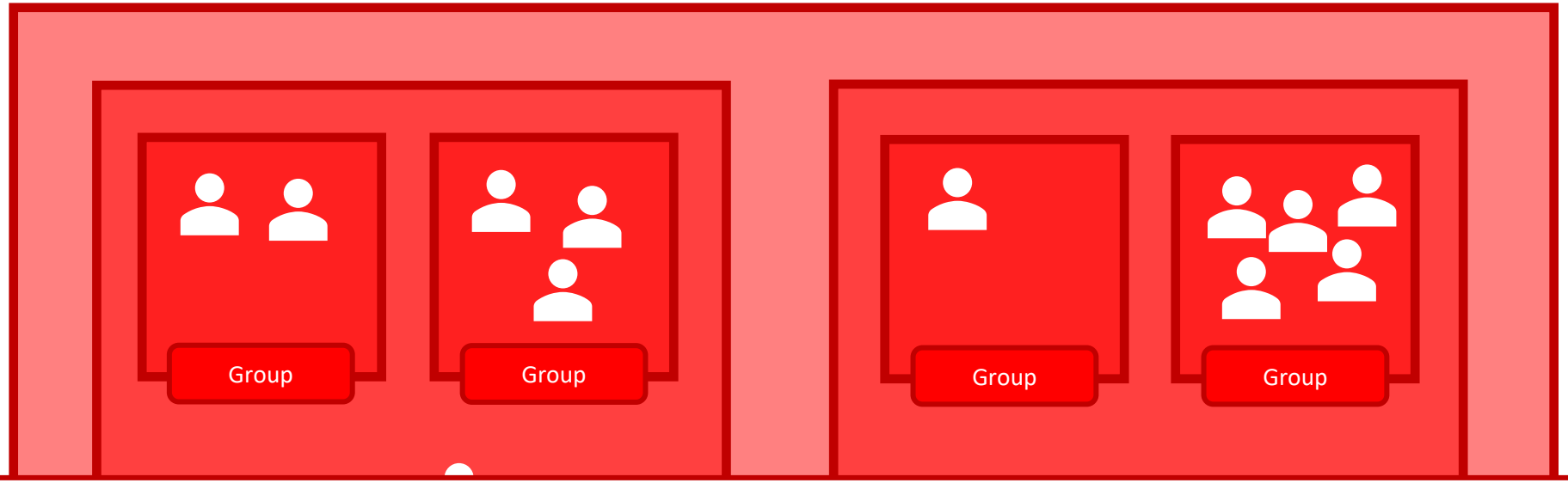
- Stai già usando SignalR
- Esiste un client SignalR nel linguaggio scelto
  - Clients supportati da MS:

- Hai necessità di più protocolli di trasporto
  - WebSocket
  - Server sent events
  - Long polling

- Hai bisogno solo di WebSocket
- Subprotocollo custom over WebSocket
  - MQTT over WebSocket
  - AMQP over WebSocket
- È sufficiente un'infrastruttura «lite» per la tua applicazione

INSIDE AZURE WEB PUBSUB





Client URL Generator

Hub *
Hub

User ID
User ID

Token Lifetime (Minutes) *
60

Select Sign Key
Primary Secondary

Roles

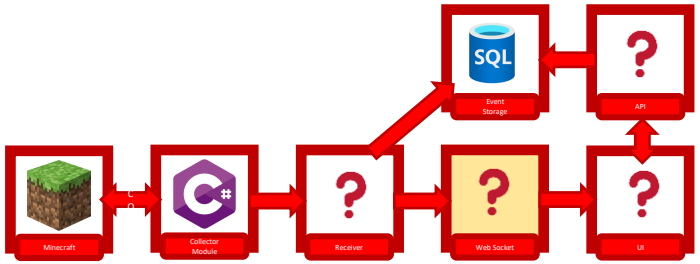
- [x] Send To Groups
- (•) Allow Sending To All Groups
- ( ) Allow Sending To Specific Groups

- [x] Join/Leave Groups
- (•) Allow Joining/Leaving All Groups
- ( ) Allow Joining/Leaving Specific Groups

Client Access URL

wss://xmasdev-events.webpubsub.azure.com/client/hubs/Hub?access_token=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJhdWQiOiJ3c3M6Ly94bWFzZGV2LWV2ZW50cy53ZWJwdWJzdWIu...

SENDING DATA WITH AZURE WEB PUBSUB



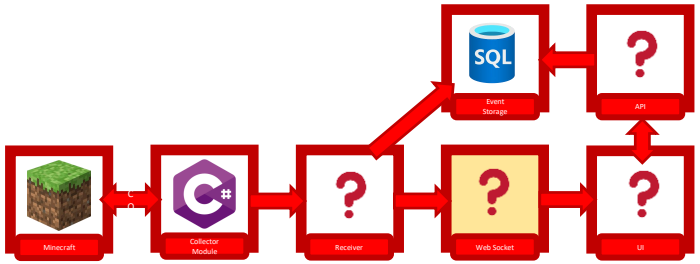
- ServiceClient C# Azure.Messaging.WebPubSub
- AzFunction Bindings Microsoft.Azure.WebJobs.Extensions.WebPubSub

```
[FunctionName(nameof(EventNotifier))]
public static async Task NotifyEventAsync(
    [ServiceBusTrigger("%TopicName%", "%SubscriptionName%", Connection = "ServiceBusConnectionString")] string rawpayload
    , [WebPubSub(Hub = "alarms", Connection = "WebPubSubConnectionString")] IAsyncCollector<WebPubSubAction> alarms
    , ILogger log
)
{
    var payload = Payloads.DeserializePayload(rawpayload);

    if (payload is null)
        return;

    if (payload is AlarmPayload)
    {
        await alarms.AddAsync(WebPubSubAction.CreateSendToAllAction(rawpayload));
        await alarms.FlushAsync();
    }
}
```

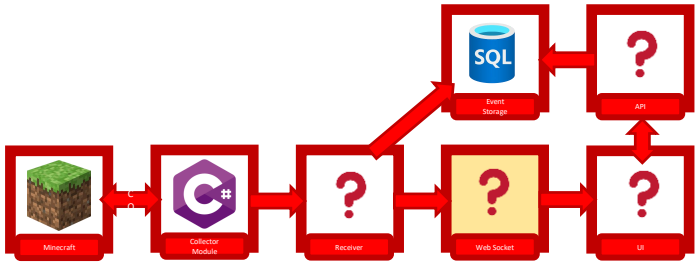
AZURE WEB PUBSUB PRICING



Pricing tier	Free For individual dev/test	Standard For production workloads	Premium(Preview) For production workloads with more supported features
Features			
Connections	Up to 20 connections	1,000 connections/unit	1,000 connections/unit
Included Messages	20,000/Day	1,000,000/Unit/Day	1,000,000/Unit/Day
Additional Messages	-	Unlimited	Unlimited
SLA	-	99.9%	99.95%
SSL	✓	✓	✓
Autoscale ⓘ	-	-	✓
Availability Zone ⓘ	-	-	✓ (In supported regions)
Custom Domains	-	-	✓
Pricing			
Estimated Price	-	41.41 EUR/Month/Unit	51.44 EUR/Month/Unit
Additional Message Costs ⓘ	-	0.84 EUR per million messages	0.84 EUR per million messages

MONITORING PLATFORM – WEB SOCKET

Web Pub Sub o SignalR



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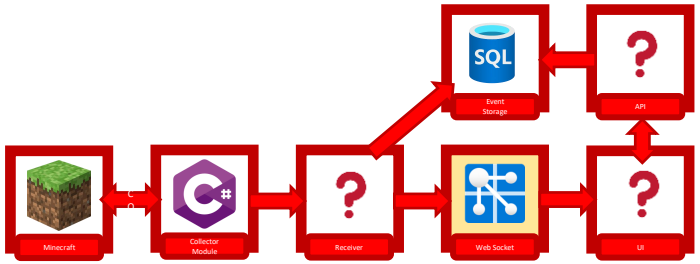
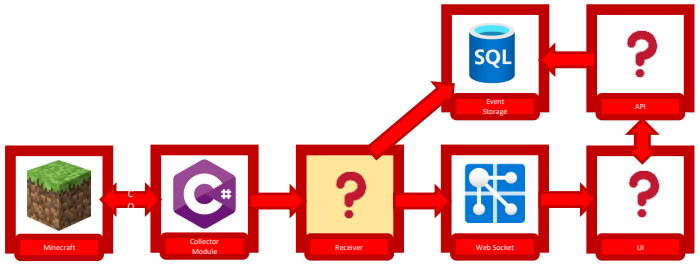


MONITORING PLATFORM - RECEIVER



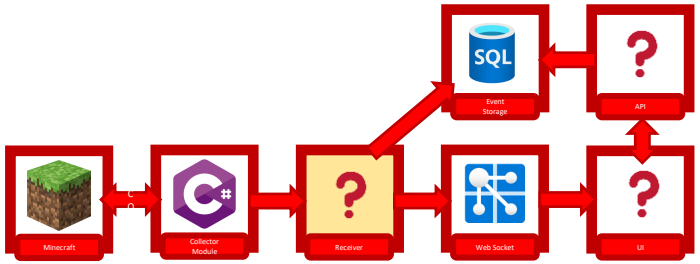
- Deve ricevere degli eventi sporadici dall'impianto produttivo
- Deve persistere questi eventi sull'Event Storage
- Deve notificare via WebSocket ogni evento

```
[FunctionName("ReceiverFunction")]
public static async Task<IActionResult> Run(
    [HttpTrigger(AuthorizationLevel.Function, "post", Route = null)] HttpRequest req,
    [WebPubSub(Hub = "CookieFactoryEvents", Connection = "WebPubSubConnectionString")] IAsyncCollector<WebPubSubAction> pubsub,
    [Sql("CookieFactoryEvents", "SqlConnectionString")] IAsyncCollector<CookieFactoryEvent> database,
    ILogger log)
{
    string requestBody = await new StreamReader(req.Body).ReadToEndAsync();
    var cloudEvent = JsonSerializer.Deserialize<CloudEvent>(requestBody);
    await pubsub.AddAsync(WebPubSubAction.CreateSendToAllAction(requestBody));
    await pubsub.FlushAsync();

    var cookieFactoryEvent = cloudEvent.Data.Deserialize<CookieFactoryEvent>();
    await database.AddAsync(cookieFactoryEvent);
    await database.FlushAsync();

    return new OkResult();
}
```

AZURE FUNCTIONS FREE GRANT



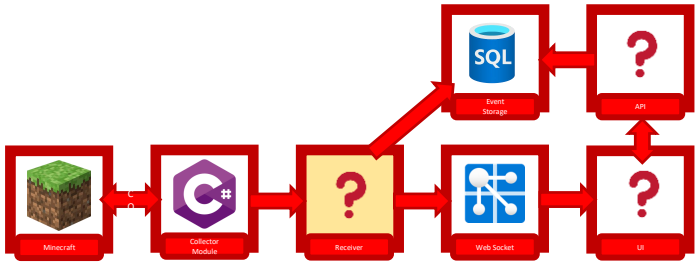
- Solamente con **Consumption Plan** in sottoscrizioni **Pay As You Go**
- Fino a 400.000 GBs ed 1 milioni di invocazioni

- Oppure le posso hostare in un AppServicePlan free

- Nonostante abbiano un free grant, le Azure Functions richiedono sempre lo **Storage Account** per funzionare (che non è gratuito)

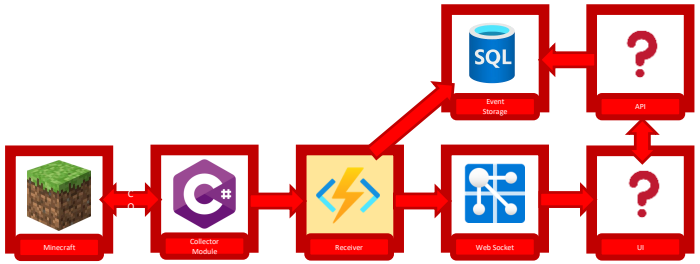
MONITORING PLATFORM - RECEIVER

Azure Functions

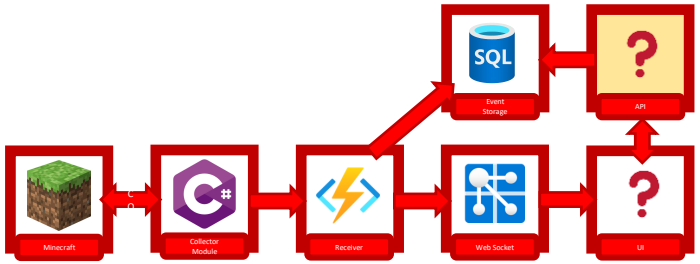


MONITORING PLATFORM - RECEIVER

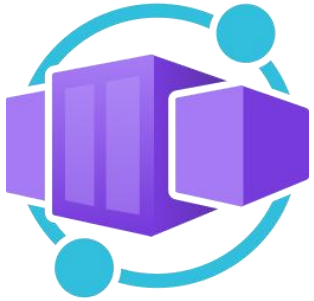
Azure Functions



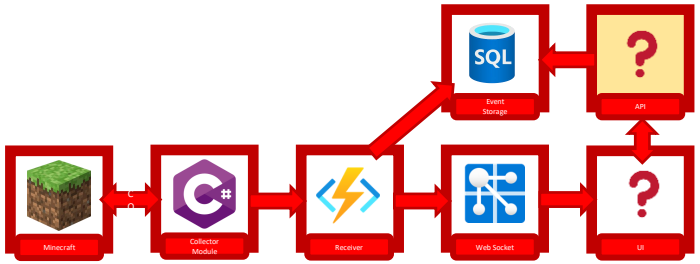
MONITORING PLATFORM - API



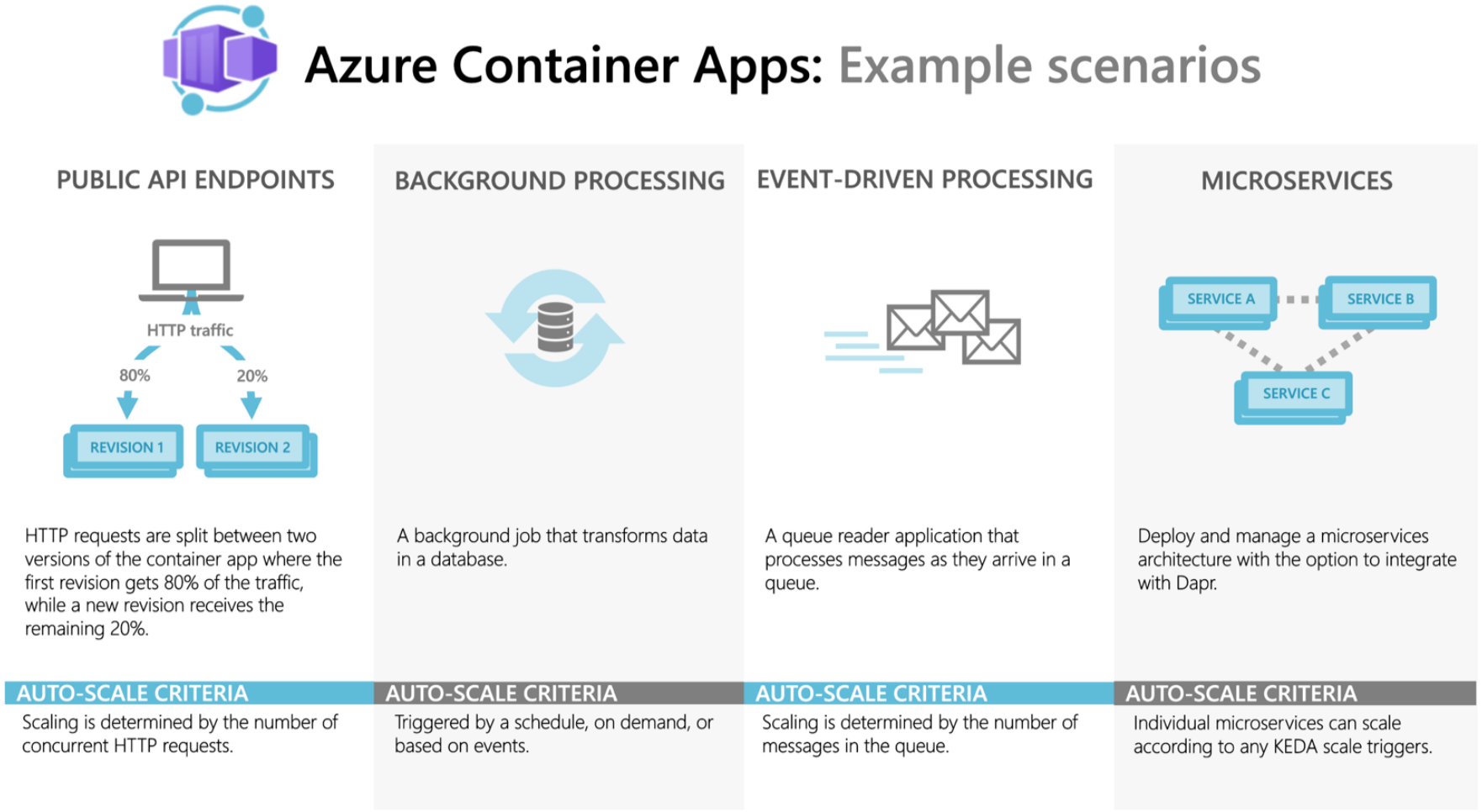
- Deve leggere gli eventi dall'Event Storage
- Deve fare delle piccole aggregate
- Deve esporre i contenuti via REST per renderli disponibili alla UI



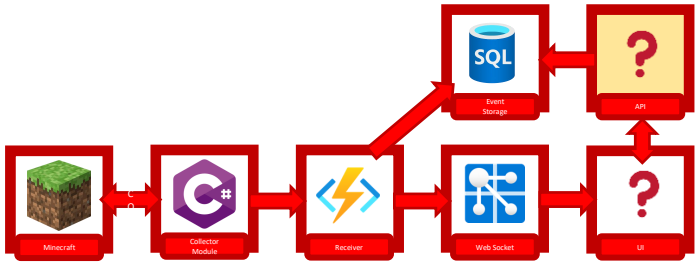
AZURE CONTAINER APPS



Piattaforma serverless per l'esecuzione di applicazioni containerizzate



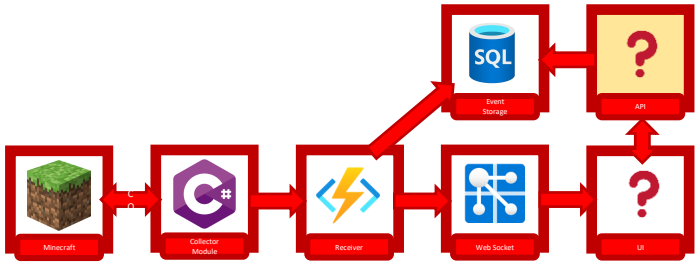
AZURE CONTAINER APPS FREE GRANT



- Ogni mese, gratis:
  - Fino a 180,000 vCPU-seconds
  - 360,000 GiB-seconds
  - 2 million di richieste

- Altre opzioni per ~~non pagarle~~ pagare di meno:
  - Scalare a 0 il numero di repliche
  - Quando non ci sono richieste HTTP in ingresso ed il numero di repliche è al minimo, viene applicato un prezzario ridotto

AZURE APP SERVICE WEB APPS



Servizio gestito per hostare applicazioni web e REST API

Vasta scelta di linguaggi utilizzabili

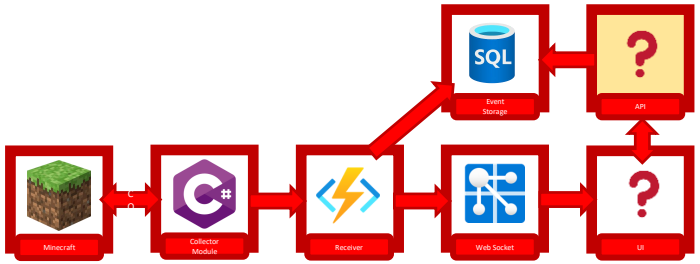
Supporta container docker

Aggiornamenti e patch di sicurezza gestite

Supporto all'autenticazione senza modificare il codice applicativo

Gestione del CORS out of the box

AZURE APP SERVICE PLAN FREE TIER

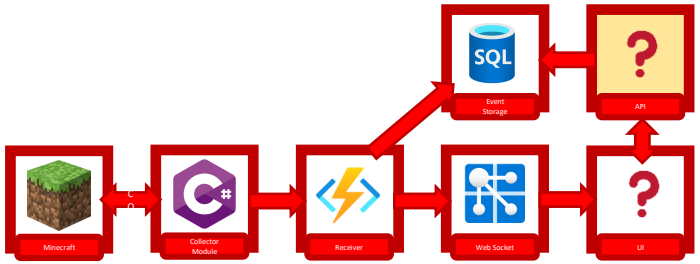


Posso avere solo un'istanza di App Service Plan Free per sottoscrizione

	Name	ACU/vCPU	vCPU	Memory (GB)	Remote Storage (GB)	Scale (instance)	Cost per hour (instance)	Cost per month (instance)
	Dev/Test (For less demanding workloads)							
☑	Free F1	60 minutes/day...	N/A	1	1	N/A	**Free**	**Free**
	Basic B1	100	1	1.75	10	3	**0.017 EUR**	**12.30 EUR**
	Basic B2	100	2	3.5	10	3	**0.034 EUR**	**24.60 EUR**
	Basic B3	100	4	7	10	3	**0.066 EUR**	**48.516 EUR**
	Production (For most production workloads)							
	Premium v3 P0V3	195*	1	4	250	30	**0.079 EUR**	**57.809 EUR**
	Premium v3 P1V3	195	2	8	250	30	**0.127 EUR**	**92.932 EUR**
	Premium v3 P2V3	195	4	16	250	30	**0.255 EUR**	**185.864 EUR**
	Premium v3 P3V3	195	8	32	250	30	**0.509 EUR**	**371.728 EUR**
	Standard S1	100	1	1.75	50	10	**0.089 EUR**	**64.915 EUR**
	Standard S2	100	2	3.5	50	10	**0.178 EUR**	**129.831 EUR**
	Standard S3	100	4	7	50	10	**0.356 EUR**	**259.662 EUR**

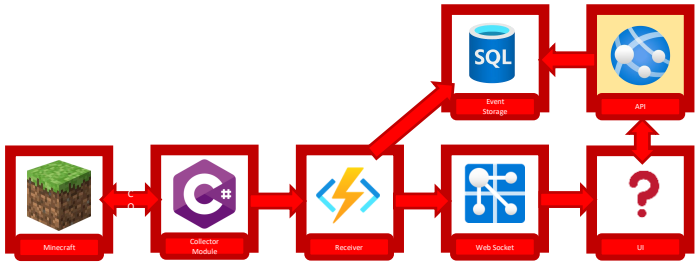
MONITORING PLATFORM - API

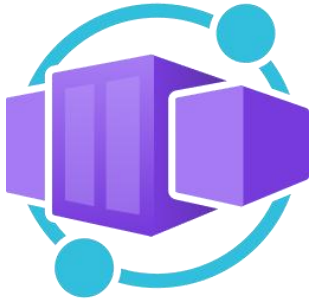
Container Apps o Web Apps



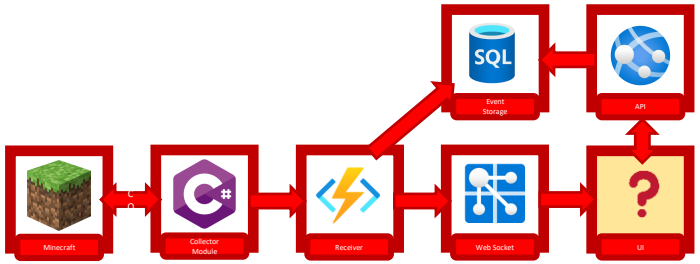
MONITORING PLATFORM - API

Container Apps o Web Apps

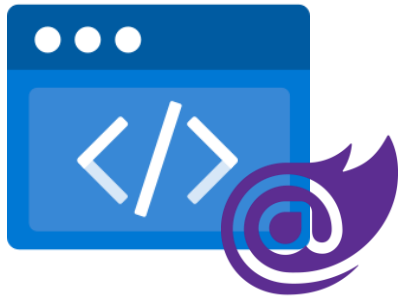




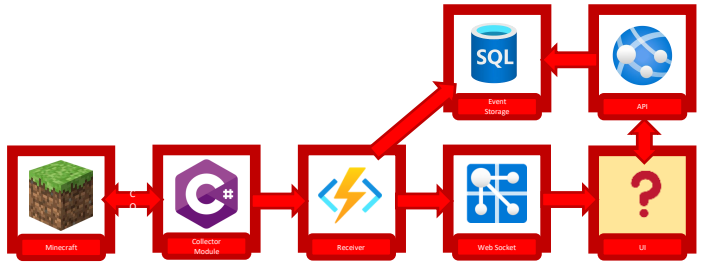
MONITORING PLATFORM - UI



- Applicativo per la consultazione degli eventi storici e la ricezione delle notifiche a partire dall'impianto produttivo
- Non ha necessità di un backend applicativo. Si può appoggiare alle API già implementate
- Può essere un'app statica



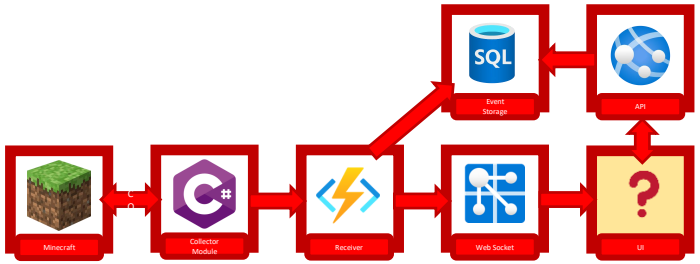
AZURE STATIC WEBAPP



È un servizio che consente di creare applicazioni Web statiche.

- Pubblicazione di contenuti statici con HTML, CSS e JavaScript
- Supporto alle API
- CI/CD con GitHub and Azure DevOps
- Contenuto statico distribuito globalmente
- Certificato SSL gratuito
- Gestione di ambiente di Staging

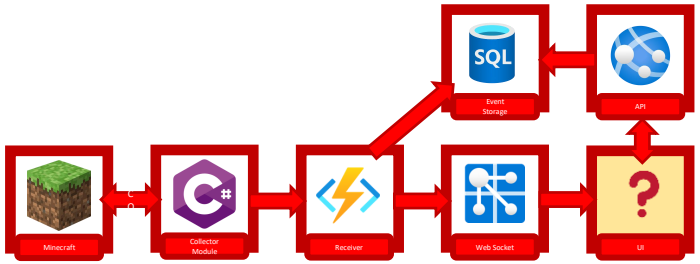
AZURE BLOB STATIC WEBSITE VS AZURE STATIC WEBAPP

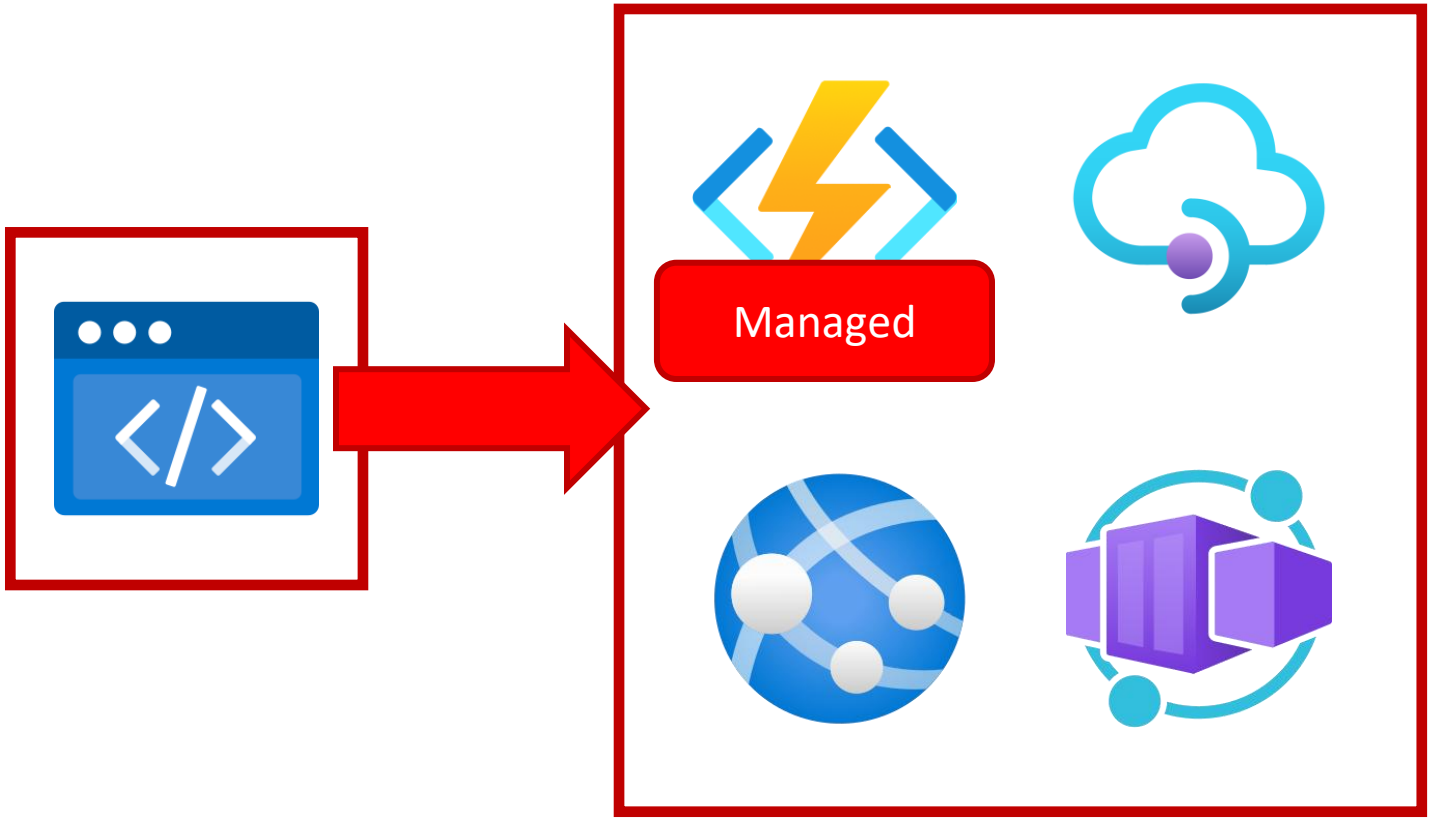


- ✔ SSL
- 😟 CI/CD non gestito
- ❌ No Multiregion
- ✔ Custom Domain
- ❌ No HTTPS con Custom Domains

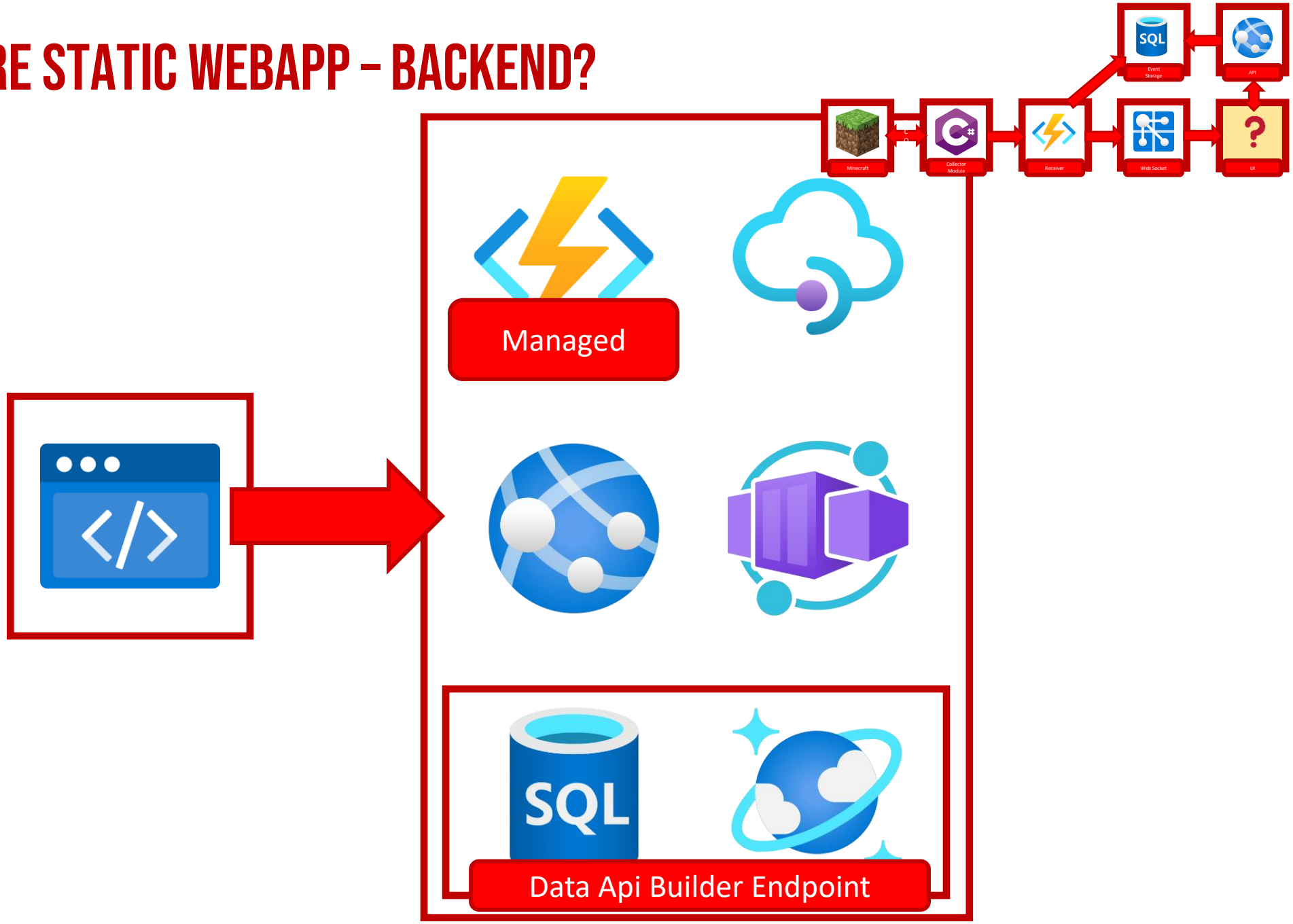
- ✔ SSL
- ✔ CI/CD integrate
- ✔ Globalmente distribuito
- ✔ Custom Domain
- ✔ HTTPS con Custom Domain

AZURE STATIC WEBAPP – BACKEND?

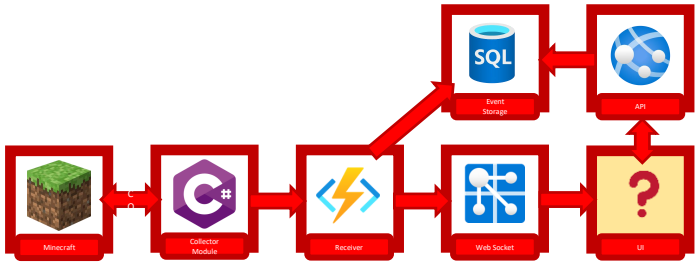


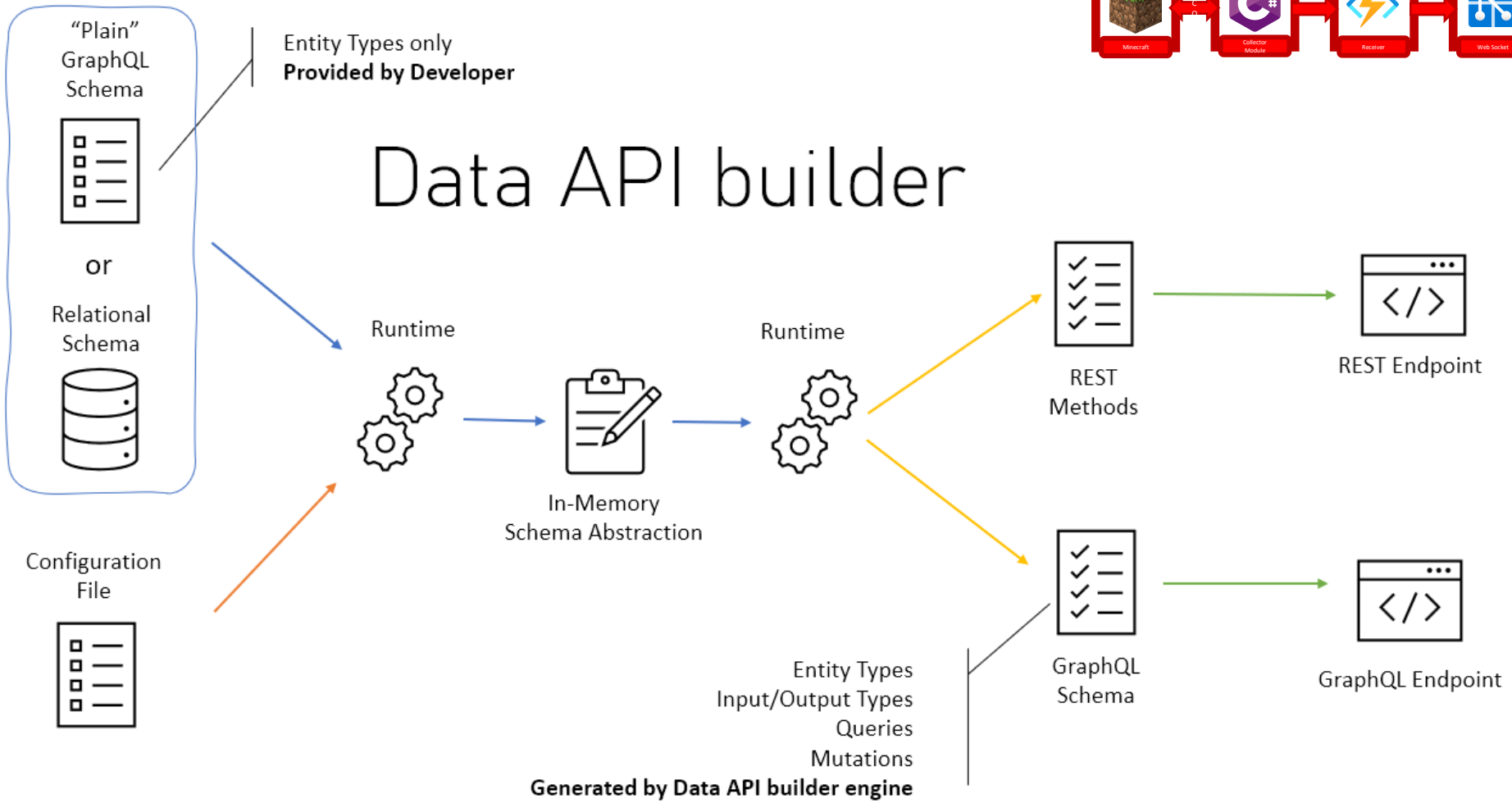


AZURE STATIC WEBAPP – BACKEND?



DATA API BUILDER?

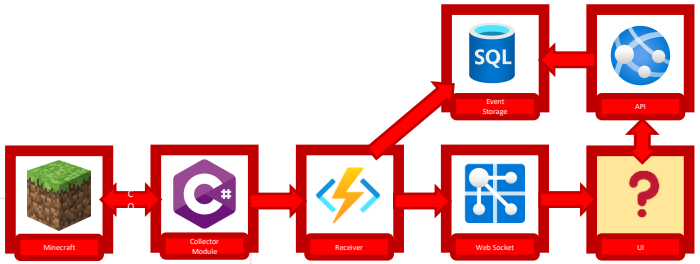




Per utilizzare DAB in locale

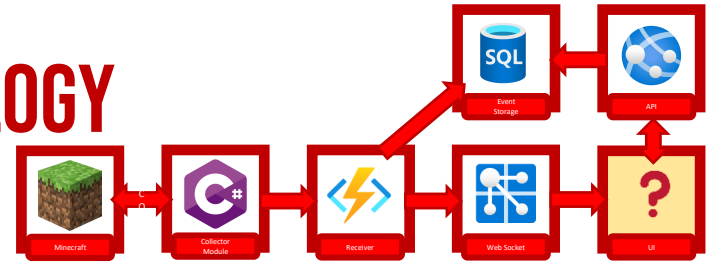
```
dotnet tool install -g Microsoft.DataApiBuilder
```

AZURE STATIC WEBAPP PRICING



Plan/Features	**Free** For hobby or personal projects	**Standard** For general purpose production apps
Price	Free	8.42 EUR per app per month
Included bandwidth	100 GB per subscription	100 GB per subscription
Bandwidth overage	Free	0.19 EUR per GB per subscription
Custom domains	2 per app	5 per app
SSL certificates	Free	Free
Custom authentication	-	✓
Private endpoints	-	✓
Max app size	250 MB	500 MB
Staging environments	3	10
Azure functions	Managed	Managed or Bring your own
Enterprise-grade edge	-	16.40 EUR per app per month

REQUIRED TOOLS TO WORK WITH THIS TECHNOLOGY



Static Web App Command line tool (swa)

```
npm install -g @azure/static-web-apps-cli
```

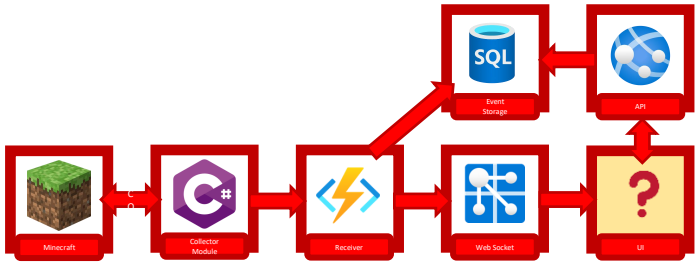
Wasm tools to output Blazor wasm

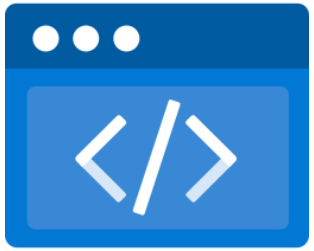
```
dotnet workload install -g wasm-tools-net6
```

https://azure.github.io/static-web-apps-cli/docs/use/install/

MONITORING PLATFORM - UI

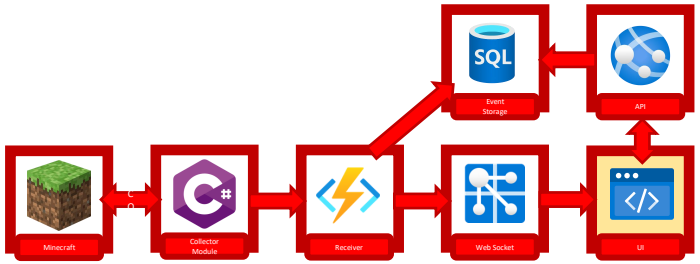
Azure Static Web App o Azure Static Web App?



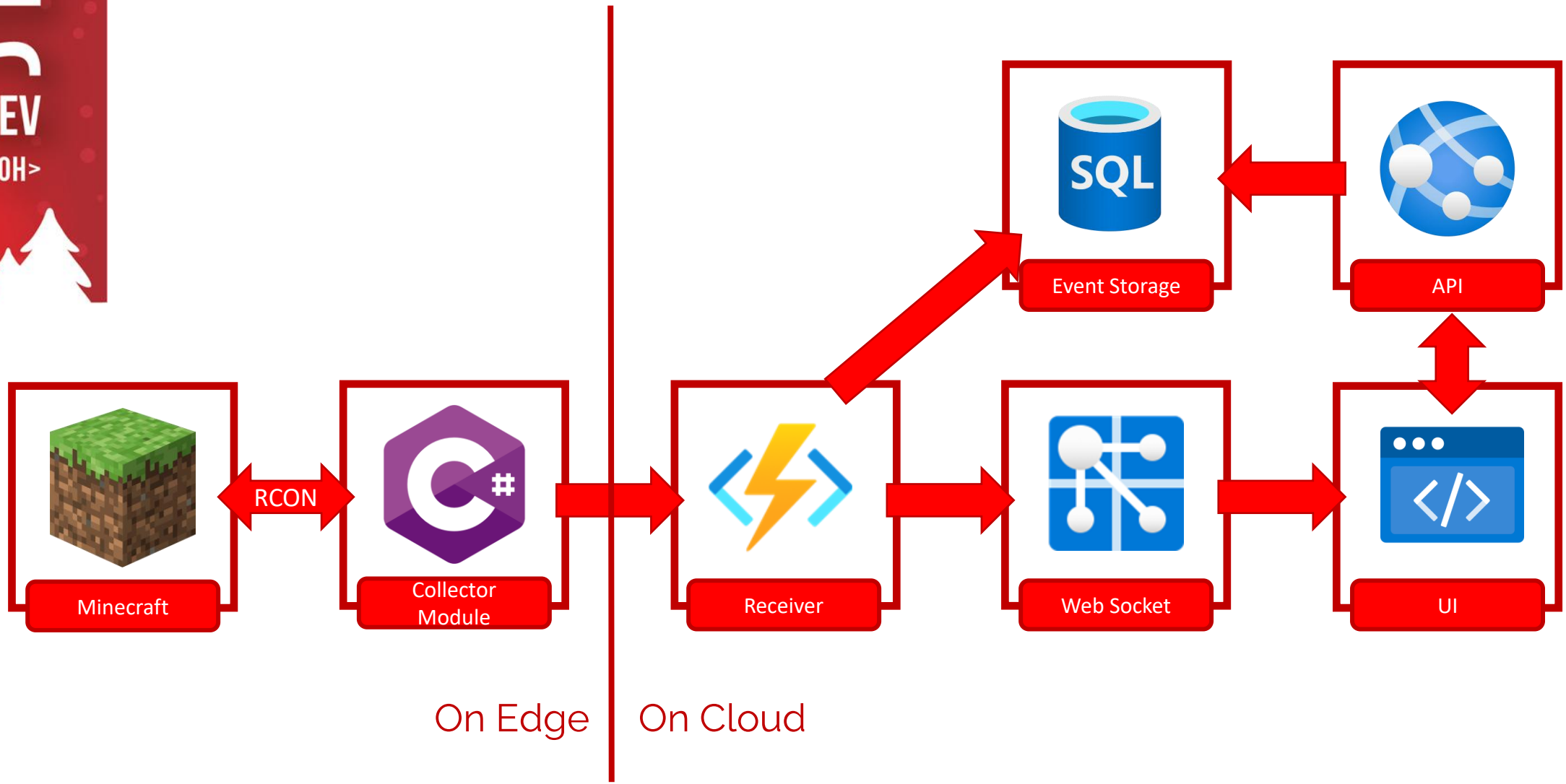


MONITORING PLATFORM - UI

Azure Static Web App o Azure Static Web App?



MONITORING PLATFORM OVERVIEW



XMASDEV

<oh>oh</oh>

DEMO

ABOUT ME

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Cloud Solutions Engineer | Technical Team Leader

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Let's Connect!

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github.com/nicolaparo

XMASDEV
<OH>OH</OH>

GRAZIE!

Microsoft

avanade

BC Soft
consulenza sistemi informativi

UNIKEY
Bringing IT knowledge to the people

Reti
Società Benefit

PHILMARK
GROUP

JET BRAINS

Packt›

stickermule

NET CODE

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DotNet Toscana

Valuta questa sessione